Northern supplement to run J: 00:11 hand-off report

Core filter, early-flight family 'waypoints'. Built 19 October 2026 from the staged files on the Mac.

What was done

Run J carried few independent paths north of 30°S. Sixteen new runs put more particles in the magnetic autopilot modes. Each run also wrote a second 00:11 hand-off that takes more samples north of 34.5°S. Every sample carries an exact weight correction, so the target posterior does not change. End of flight combines these samples with run J by the balance heuristic (rule B).

Main results

1. The 00:11 latitude distribution of run J and the supplement together stays within the run-to-run spread of run J alone (Figure 1). Band masses, run J alone / combined: 30–25°S 0.0184 / 0.0194; 25–20°S 0.0023 / 0.0026.

2. The effective number of distinct 00:11 states rises from 3,987 to 13,746 in 30–25°S and from 518 to 1,049 in 25–20°S (Figure 2).

3. Moving particles between modes changes the evidence noise of each mode as 1/sqrt(particles) predicts, for all five modes within the 95 % intervals (Figure 3). The noise of the mode probabilities is therefore mainly a particle-number effect.

4. Every corrected estimate agrees with the uncorrected run: all |z| below 1.6 (Table 1). The weights add up to 1 and the program gives the same bytes as run J with the new setting off (Table 2). The 00:19 result agrees with run J (Table 3).

Limits

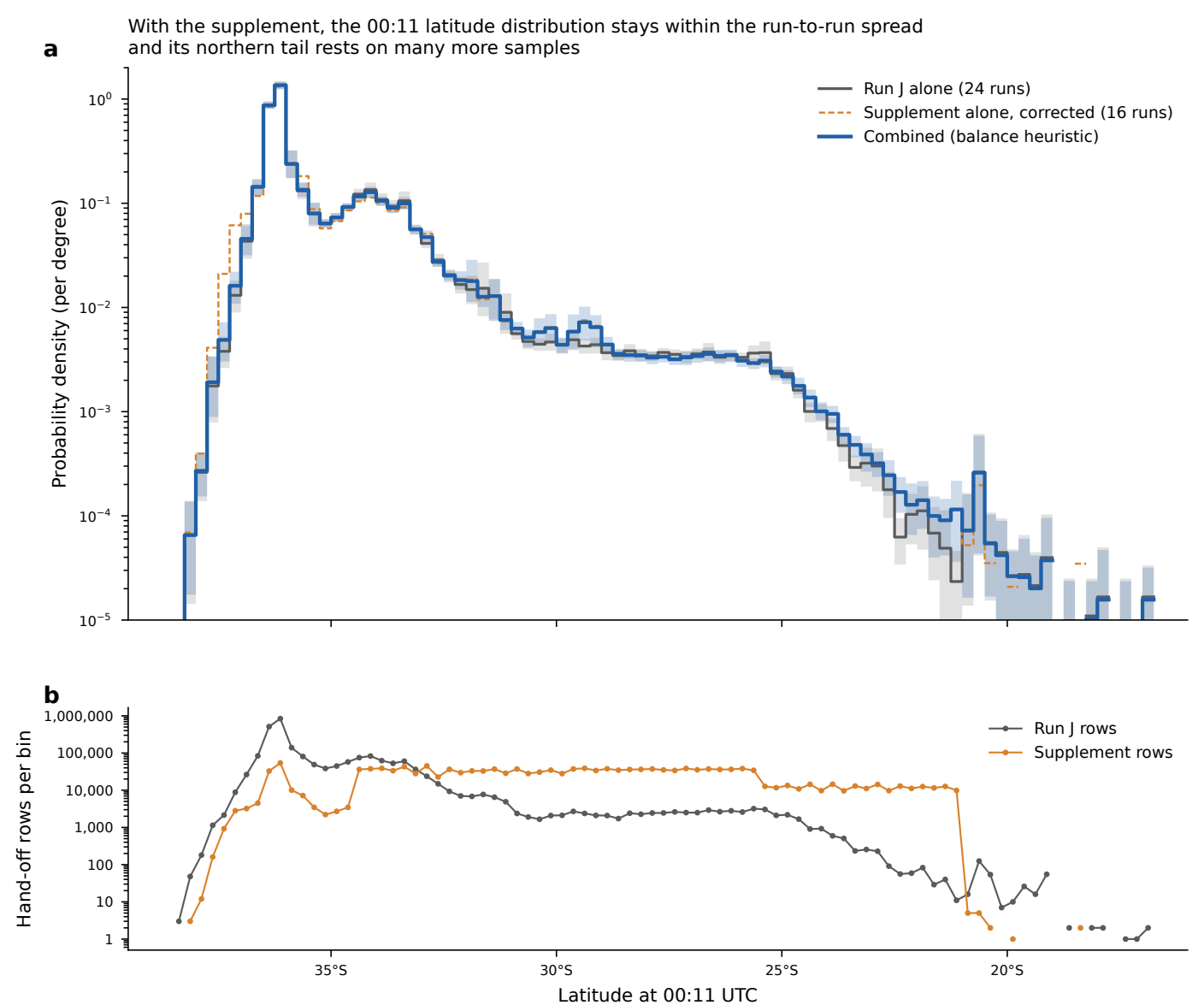
The 1,000-sample target applies to impact latitude under the 00:19 data options. Core can only count at 00:11; end of flight measures the target on impacts. Under rule B the northern mass rests mainly on the 16 supplement runs, so its interval does not shrink. Only the 'waypoints' family has a supplement. Run J convergence: PASS by project ruling (Pete, 16 October 2026); this report does not change it.

Contents

Figure 1 Latitude at 00:11. Figure 2 Distinct states by band and mode. Figure 3 Evidence noise by mode. Table 1 Prior recovery. Table 2 Correction checks. Table 3 Settings and results against run J and Davey et al. (2016), Fig. 10.3 (vector-recovered curve). The standard run report for the 16 runs is a separate file.

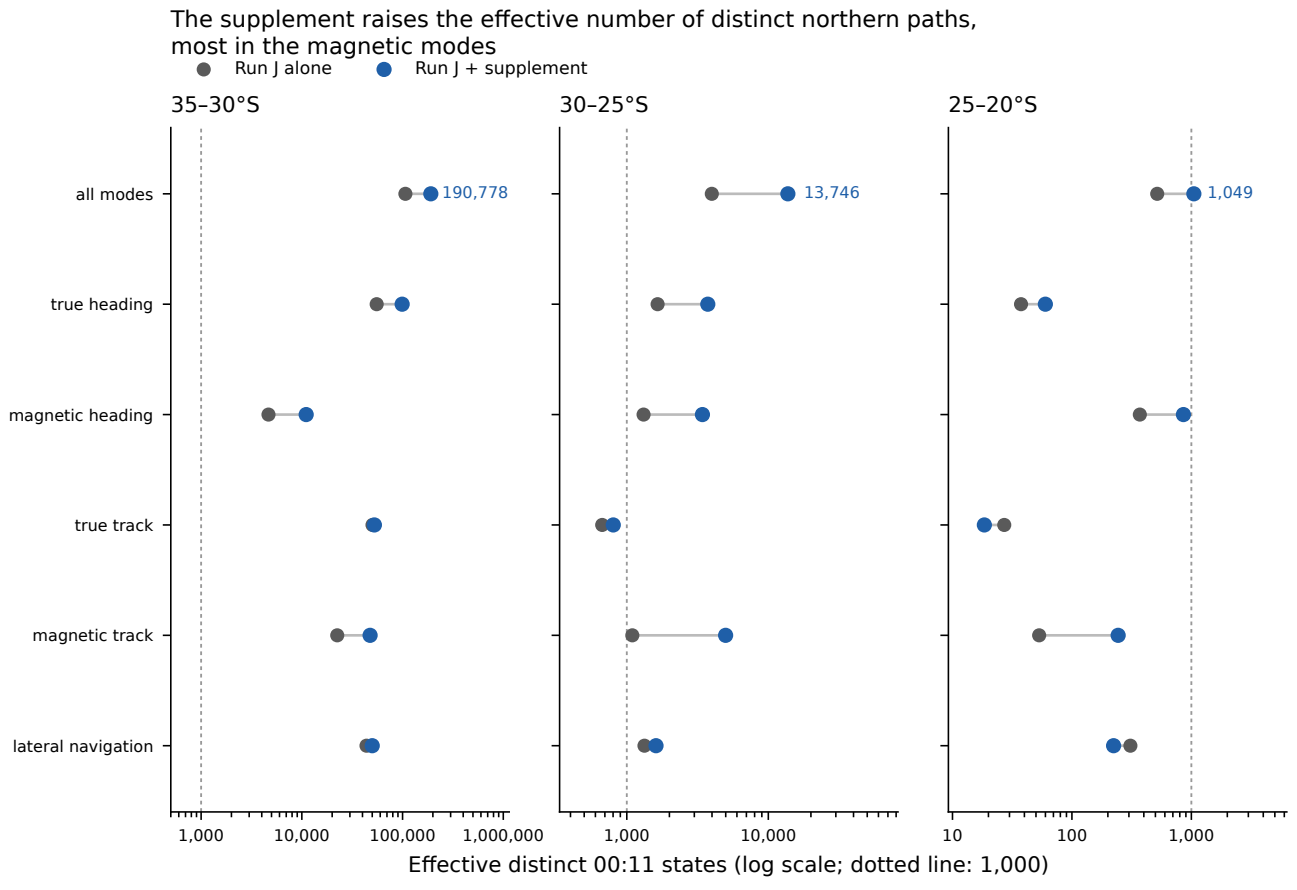
Plain: This page gives the summary of the report. The source is the core filter: run J, early-flight family 'waypoints' (24 runs), and the northern supplement (16 new runs). The prior track starts from the 18:01 radar position and follows a random route through the 2014 waypoints. The 00:11 data are used; the 00:19 data are not. The one-engine physics and the fuel-flow factor are provisional. No search areas, ocean drift or end-of-flight data are applied. The 18:25:27 signal is not used. The other three early-flight families are not included.

Technical: Values quoted are those of the figures and tables that follow; sources are given in their footnotes. Core 00:11 hand-offs (filtering distribution at m0011, look-ahead correction exp(log_correction) applied): run J waypoints seeds 1-24 (tag run-j, 1244b7c; 100k rows per seed) and supplement seeds 1001-1016 (rev 3bf0801; particles per mode [TH, MH, TT, MT, LN] = [1.2, 1.3, 1.1, 2.2, 1.2]M vs run J [1.0, 0.5, 2.5, 0.5, 2.5]M; supplement hand-off 100k rows per seed, tilt r(lat) = 0.1 + 0.9 h/H over 00:11 latitude -34.5..-21.0). BFO sd 4.3177 Hz; kappa_x ~ N(1.0, 0.046) and one-engine s7 + s8 PROVISIONAL; 18:25:27 R600 held out. Search case: none; ocean models: none; end-of-flight 00:19 options: not applied. Combination rule B: balance heuristic with r_hat (r_hat.csv), n1 = 2,399,999, n2 = 1,600,003 rows.



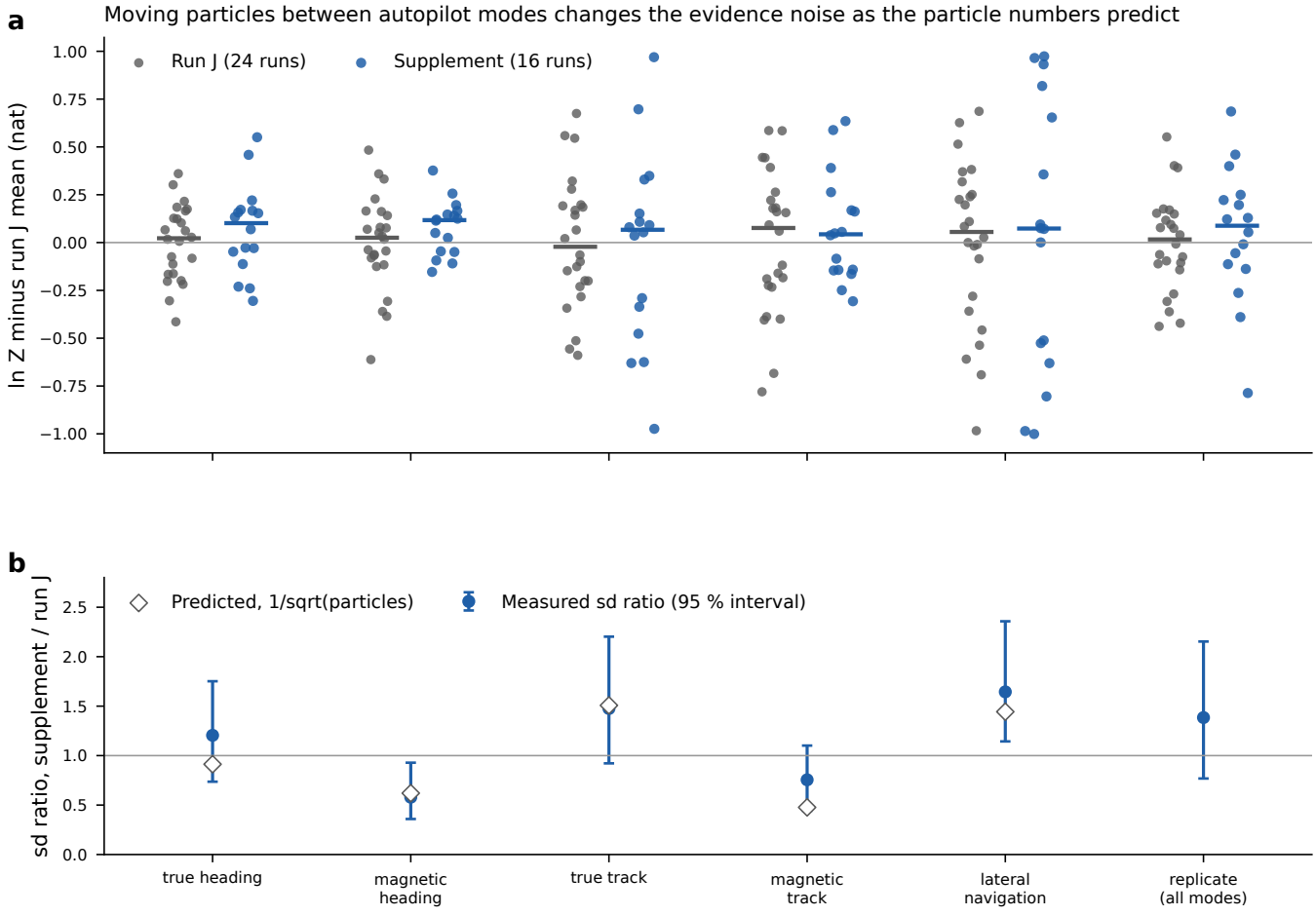
Plain: The curves show where the aircraft is at 00:11, by latitude, on a log scale. The combined curve uses run J and the supplement together. It agrees with run J alone. North of 30 S it rests on many more samples. The shaded bands show the spread between runs (5 to 95 %). The source is the core filter: run J, early-flight family 'waypoints' (24 runs), and the northern supplement (16 new runs). The prior track starts from the 18:01 radar position and follows a random route through the 2014 waypoints. The 00:11 data are used; the 00:19 data are not. The one-engine physics and the fuel-flow factor are provisional. No search areas, ocean drift or end-of-flight data are applied. The 18:25:27 signal is not used. The other three early-flight families are not included.

Technical: Density per degree in 0.25 deg bins of 00:11 latitude. Bands: seed bootstrap (400 resamples of production and supplement seeds, both resampled), 5-95 %. Panel b: hand-off rows per bin (sample count, not probability). Band masses (production / combined, 5-95 %): 35-30 S 0.241 [0.217-0.267] / 0.237 [0.217-0.256]; 30-25 S 0.0184 [0.0165-0.0204] / 0.0194 [0.0164-0.0219]; 25-20 S 0.0023 [0.0019-0.0026] / 0.0026 [0.0022-0.0030]. The northern mass interval does not shrink: under rule B it rests mainly on the 16 supplement seeds' mode probabilities. The supplement alone is noisy south of 35 S, where it holds few rows (defensive share 0.1 only). Core 00:11 hand-offs (filtering distribution at m0011, look-ahead correction exp(log_correction) applied): run J waypoints seeds 1-24 (tag run-j, 1244b7c; 100k rows per seed) and supplement seeds 1001-1016 (rev 3bf0801; particles per mode [TH, MH, TT, MT, LN] = [1.2, 1.3, 1.1, 2.2, 1.2]M vs run J [1.0, 0.5, 2.5, 0.5, 2.5]M; supplement hand-off 100k rows per seed, tilt r(lat) = 0.1 + 0.9 h/H over 00:11 latitude -34.5..-21.0). BFO sd 4.3177 Hz; kappa_x ~ N(1.0, 0.046) and one-engine s7 + s8 PROVISIONAL; 18:25:27 R600 held out. Search case: none; ocean models: none; end-of-flight 00:19 options: not applied. Combination rule B: balance heuristic with r_hat (r_hat.csv), n1 = 2,399,999, n2 = 1,600,003 rows.



Plain: Each dot gives the effective number of different aircraft states at 00:11 in one latitude band. Copies of one state count once. Grey is run J alone. Blue is run J and the supplement together. The supplement gives about 3 times more in 30 to 25 S and 2 times more in 25 to 20 S. The source is the core filter: run J, early-flight family 'waypoints' (24 runs), and the northern supplement (16 new runs). The prior track starts from the 18:01 radar position and follows a random route through the 2014 waypoints. The 00:11 data are used; the 00:19 data are not. The one-engine physics and the fuel-flow factor are provisional. No search areas, ocean drift or end-of-flight data are applied. The 18:25:27 signal is not used. The other three early-flight families are not included.

Technical: Kish ESS over distinct (seed, state) pairs, state = (latitude, longitude, altitude, BFO bias mean) of the 00:11 hand-off row; posterior weights (rule B for combined, w exp(lc)/24 for run J alone). Bands of 00:11 latitude (about 1 deg north of the impact band). Distinct states production -> combined: 35-30 S 461k -> 825k; 30-25 S 15.4k -> 90.0k; 25-20 S 1,966 -> 13,705. Under D2-D4 the effective count is lower (it depends on the 00:19 likelihood); end of flight measures it on impacts. Core 00:11 hand-offs (filtering distribution at m0011, look-ahead correction exp(log_correction) applied): run J waypoints seeds 1-24 (tag run-j, 1244b7c; 100k rows per seed) and supplement seeds 1001-1016 (rev 3bf0801; particles per mode [TH, MH, TT, MT, LN] = [1.2, 1.3, 1.1, 2.2, 1.2]M vs run J [1.0, 0.5, 2.5, 0.5, 2.5]M; supplement hand-off 100k rows per seed, tilt r(lat) = 0.1 + 0.9 h/H over 00:11 latitude -34.5..-21.0). BFO sd 4.3177 Hz; kappa_x ~ N(1.0, 0.046) and one-engine s7 + s8 PROVISIONAL; 18:25:27 R600 held out. Search case: none; ocean models: none; end-of-flight 00:19 options: not applied. Combination rule B: balance heuristic with r_hat (r_hat.csv), n1 = 2,399,999, n2 = 1,600,003 rows.



Plain: Panel a shows the evidence of each autopilot mode in each run, measured from the run J mean. The supplement runs have more particles in the magnetic modes and fewer in the true-track and lateral-navigation modes. Panel b compares the measured spread with the spread that the particle numbers predict. For most modes they agree, so the random error of the mode probabilities comes mainly from the particle number. The source is the core filter: run J, early-flight family 'waypoints' (24 runs), and the northern supplement (16 new runs). The prior track starts from the 18:01 radar position and follows a random route through the 2014 waypoints. The 00:11 data are used; the 00:19 data are not. The one-engine physics and the fuel-flow factor are provisional. No search areas, ocean drift or end-of-flight data are applied. The 18:25:27 signal is not used. The other three early-flight families are not included.

Technical: ln Z per mode = log evidence of each mode filter (diagnostics.json), run J waypoints seeds 1-24 against supplement seeds 1001-1016; 'replicate' = ln Z of the mode mixture. Panel b: ratio of seed sd (supplement / run J) with 95 % bootstrap interval (4,000 resamples of seeds), against sqrt(N_run J / N_supplement) for N particles per mode, the ratio expected if the sd falls as 1/sqrt(N). Mean ln Z per mode agrees with run J within 0.1 nat (target unchanged). Core 00:11 hand-offs (filtering distribution at m0011, look-ahead correction exp(log_correction) applied): run J waypoints seeds 1-24 (tag run-j, 1244b7c; 100k rows per seed) and supplement seeds 1001-1016 (rev 3bf0801; particles per mode [TH, MH, TT, MT, LN] = [1.2, 1.3, 1.1, 2.2, 1.2]M vs run J [1.0, 0.5, 2.5, 0.5, 2.5]M; supplement hand-off 100k rows per seed, tilt r(lat) = 0.1 + 0.9 h/H over 00:11 latitude -34.5..-21.0). BFO sd 4.3177 Hz; kappa_x ~ N(1.0, 0.046) and one-engine s7 + s8 PROVISIONAL; 18:25:27 R600 held out. Search case: none; ocean models: none; end-of-flight 00:19 options: not applied. Combination rule B: balance heuristic with r_hat (r_hat.csv), n1 = 2,399,999, n2 = 1,600,003 rows.

Prior recovery: every corrected estimate agrees with the uncorrected run (all |z| below 1.6)

Quantity	Same population (run J seed 1)	Supplement, new seeds vs run J	New seeds, normal output vs run J	Combined (rule B) vs run J
Mean latitude	+0.76	-0.37	-0.40	+0.08
Mean longitude	+1.00	-0.47	-0.50	—
Mean altitude	-1.31	-0.51	-0.54	—
Mean Mach	-0.53	+0.56	+0.57	—
Mean BFO bias	-0.22	-0.45	-0.46	—
P(north of 33°S)	-0.00	+0.48	+0.49	+0.67
P(north of 30°S)	-0.15	+0.52	+0.48	+1.00
P(north of 25°S)	-0.34	+0.89	+0.82	+1.59
P(true heading)	+0.63	+0.54	+0.53	+0.58
P(magnetic heading)	+0.05	+0.58	+0.48	+0.67
P(true track)	-0.98	-0.66	-0.65	-0.39
P(magnetic track)	+1.35	+0.42	+0.40	+0.86
P(lateral navigation)	-0.13	+0.40	+0.41	-0.28
Median latitude at 00:19	—	—	+0.26	—
Mean latitude at 00:19	—	—	-0.77	—
P(north of 33°S) at 00:19	—	—	+0.05	—
P(north of 30°S) at 00:19	—	—	-0.36	—
P(34.5–36.5°S) at 00:19	—	—	-1.28	—

Plain: The table compares each corrected supplement estimate with the estimate that has no correction. A z value is the difference divided by its standard error. All values are less than 3 in size, so the correction does not change the result. Rows marked 00:19 use the final result, with both 00:19 BTO values and no 00:19 BFO. The source is the core filter: run J, early-flight family 'waypoints' (24 runs), and the northern supplement (16 new runs). The prior track starts from the 18:01 radar position and follows a random route through the 2014 waypoints. The 00:11 data are used; the 00:19 data are not. The one-engine physics and the fuel-flow factor are provisional. No search areas, ocean drift or end-of-flight data are applied. The 18:25:27 signal is not used. The other three early-flight families are not included.

Technical: Column 1: run J waypoints seed 1, supplement drawn from its own 00:11 population, against its normal hand-off; SE from Kish ESS over distinct states. Columns 2-3: new seeds 1001-1016 against run J seeds 1-24, seed-level z = difference of seed means / sqrt(s1^2/16 + s2^2/24). Column 4: rule B combination against run J alone, production seed SE only (conservative). 00:19 rows: final.npy of the new seeds (re-allocated modes) against run J. Pass criterion |z| < 3. Core 00:11 hand-offs (filtering distribution at m0011, look-ahead correction exp(log_correction) applied): run J waypoints seeds 1-24 (tag run-j, 1244b7c; 100k rows per seed) and supplement seeds 1001-1016 (rev 3bf0801; particles per mode [TH, MH, TT, MT, LN] = [1.2, 1.3, 1.1, 2.2, 1.2]M vs run J [1.0, 0.5, 2.5, 0.5, 2.5]M; supplement hand-off 100k rows per seed, tilt r(lat) = 0.1 + 0.9 h/H over 00:11 latitude -34.5..-21.0). BFO sd 4.3177 Hz; kappa_x ~ N(1.0, 0.046) and one-engine s7 + s8 PROVISIONAL; 18:25:27 R600 held out. Search case: none; ocean models: none; end-of-flight 00:19 options: not applied. Combination rule B: balance heuristic with r_hat (r_hat.csv), n1 = 2,399,999, n2 = 1,600,003 rows.

Correction checks: the weights are exact and the program is unchanged with the new setting off

Check	Result	Verdict
Supplement alone, sum of corrected weights per run (expected 1)	0.9915–1.0103 (mean 1.0009)	pass
Normal hand-off of the new runs, sum of corrected weights	0.9954–1.0013	pass
Combined (rule B), sum of weights over both sets	0.9973	pass
Same-run combination (rule C), P(north of 30°S) at 00:11	0.02208 against normal alone 0.02197; paired z +1.09	pass
Run J seed 1, own population: sums of corrected weights	supplement 1.001, normal 0.998	pass
Supplement rows north of 34.5°S at 00:11 (prediction 0.92)	0.920 (34.5–29.5°S 0.424, 29.5–25.5°S 0.362, 25.5–21.0°S 0.135)	as designed
Unit tests (four new: exact alone, exact combined, rho, configuration)	36 + 41 passed on deskstar	pass
Gates B and C; regress-seed1	byte-identical to 15 Oct references; −97.51251428262529 equal	pass
Flag off: run J repro seed 1 at the new revision	byte-identical to run J (8 files)	pass
Flag on: run J waypoints seed 1	all 11 normal outputs byte-identical to run J	pass
Mean ln Z per mode, new runs against run J	within 0.1 nat for every mode	pass

Plain: The table shows the checks on the weight correction. The corrected weights add up to 1 within the random error. The two ways to combine the hand-offs agree with the normal hand-off. With the new setting off, the program gives the same bytes as run J. The source is the core filter: run J, early-flight family 'waypoints' (24 runs), and the northern supplement (16 new runs). The prior track starts from the 18:01 radar position and follows a random route through the 2014 waypoints. The 00:11 data are used; the 00:19 data are not. The one-engine physics and the fuel-flow factor are provisional. No search areas, ocean drift or end-of-flight data are applied. The 18:25:27 signal is not used. The other three early-flight families are not included.

Technical: Sums: sum over rows of weight x exp(log_correction), per run. Rule B: balance heuristic across replicates with r_hat (r_hat.csv). Rule C: same-replicate balance heuristic, normal row weight / (1 + rho), supplement row weight x rho / (1 + rho), rho from production-mis.npy / mis.npy; paired over the 16 runs. Byte identity: deskstar job b6389957 (identity-flagoff-repro.txt, identity-flagon-waypoints.txt); handoff.toml compared uncompressed. Core 00:11 hand-offs (filtering distribution at m0011, look-ahead correction exp(log_correction) applied): run J waypoints seeds 1-24 (tag run-j, 1244b7c; 100k rows per seed) and supplement seeds 1001-1016 (rev 3bf0801; particles per mode [TH, MH, TT, MT, LN] = [1.2, 1.3, 1.1, 2.2, 1.2]M vs run J [1.0, 0.5, 2.5, 0.5, 2.5]M; supplement hand-off 100k rows per seed, tilt r(lat) = 0.1 + 0.9 h/H over 00:11 latitude -34.5..-21.0). BFO sd 4.3177 Hz; kappa_x ~ N(1.0, 0.046) and one-engine s7 + s8 PROVISIONAL; 18:25:27 R600 held out. Search case: none; ocean models: none; end-of-flight 00:19 options: not applied. Combination rule B: balance heuristic with r_hat (r_hat.csv), n1 = 2,399,999, n2 = 1,600,003 rows.

Settings and results: the supplement, run J and the published model

Item	Supplement (16 runs)	Run J waypoints (24 runs)	Davey et al. (2016)
Seeds	1001–1016 (16)	1–24 (24)	—
Particles per run	7,000,000	7,000,000	—
Particles per mode (TH, MH, TT, MT, LN)	1.2, 1.3, 1.1, 2.2, 1.2 M	1.0, 0.5, 2.5, 0.5, 2.5 M	—
00:11 hand-off rows per run	100,000 normal + 100,000 supplement	100,000	—
Supplement tilt	r = 0.1 + 0.9 h/H; 45/40/15 % over 34.5–29.5 / 29.5–25.5 / 25.5–21.0°S	none	—
BFO noise sd	4.3177 Hz	4.3177 Hz	4.3177 Hz (Table 5.1)
18:25:27 R600 burst	held out	held out	used
Fuel-flow factor on extrapolated steps	N(1.0, 0.046), provisional	N(1.0, 0.046), provisional	no fuel model
Engine binary	3bf0801 (run J + flag)	1244b7c (tag run-j)	—
Median latitude at 00:19	37.11°S	37.10°S	37.43°S
95 % interval at 00:19	33.25–38.02°S	33.05–37.65°S	34.83–39.21°S
P(34.5–36.5°S) at 00:19	0.140	0.167	0.270
P(north of 33°S) at 00:19	0.0232	0.0246	0.0040
ln Z (pooled)	-127.77	−127.82 (seed sd 0.26)	—
Split-half, mean of balanced partitions	0.883 (16 runs); 8-run subsets 0.837	0.934 (24 runs); 8-run subsets 0.892	—

Plain: The table compares the settings and the results of the supplement runs with run J and with the published model. The supplement uses the run J model; only the particle split between the autopilot modes and the extra 00:11 hand-off are different. The 00:19 rows use the final result, with both 00:19 BTO values and no 00:19 BFO. They agree with run J within the run-to-run spread. The source is the core filter: run J, early-flight family 'waypoints' (24 runs), and the northern supplement (16 new runs). The prior track starts from the 18:01 radar position and follows a random route through the 2014 waypoints. The 00:11 data are used; the 00:19 data are not. The one-engine physics and the fuel-flow factor are provisional. No search areas, ocean drift or end-of-flight data are applied. The 18:25:27 signal is not used. The other three early-flight families are not included.

Technical: Supplement 00:19 values from the pooled summary.json of the 16 runs (mh370 summarise over merged seeds; evidence pooling by mode). Run J values from results/run-j/README.md (24 runs). Davey et al. (2016), Fig. 10.3: vector-recovered curve (engine/.sources/davey-2016/fig10-3-bto-bfo-latitude-pdf-vector.csv, 5315fde); its quantiles and band masses computed here on a 0.01 deg grid. Split-half: mean over all balanced partitions (6,435 for 16 runs); no calibrated floor for 16 runs; run J's ruling (PASS, Pete 16 Oct 2026) is not affected. Southern 2.5 % point differs (38.02 against 37.65°S): a run-to-run effect of the 38°S side mode (Monte Carlo noise; see the standard report, Figure 1). Core 00:11 hand-offs (filtering distribution at m0011, look-ahead correction exp(log_correction) applied): run J waypoints seeds 1-24 (tag run-j, 1244b7c; 100k rows per seed) and supplement seeds 1001-1016 (rev 3bf0801; particles per mode [TH, MH, TT, MT, LN] = [1.2, 1.3, 1.1, 2.2, 1.2]M vs run J [1.0, 0.5, 2.5, 0.5, 2.5]M; supplement hand-off 100k rows per seed, tilt r(lat) = 0.1 + 0.9 h/H over 00:11 latitude -34.5..-21.0). BFO sd 4.3177 Hz; kappa_x ~ N(1.0, 0.046) and one-engine s7 + s8 PROVISIONAL; 18:25:27 R600 held out. Search case: none; ocean models: none; end-of-flight 00:19 options: not applied. Combination rule B: balance heuristic with r_hat (r_hat.csv), n1 = 2,399,999, n2 = 1,600,003 rows.